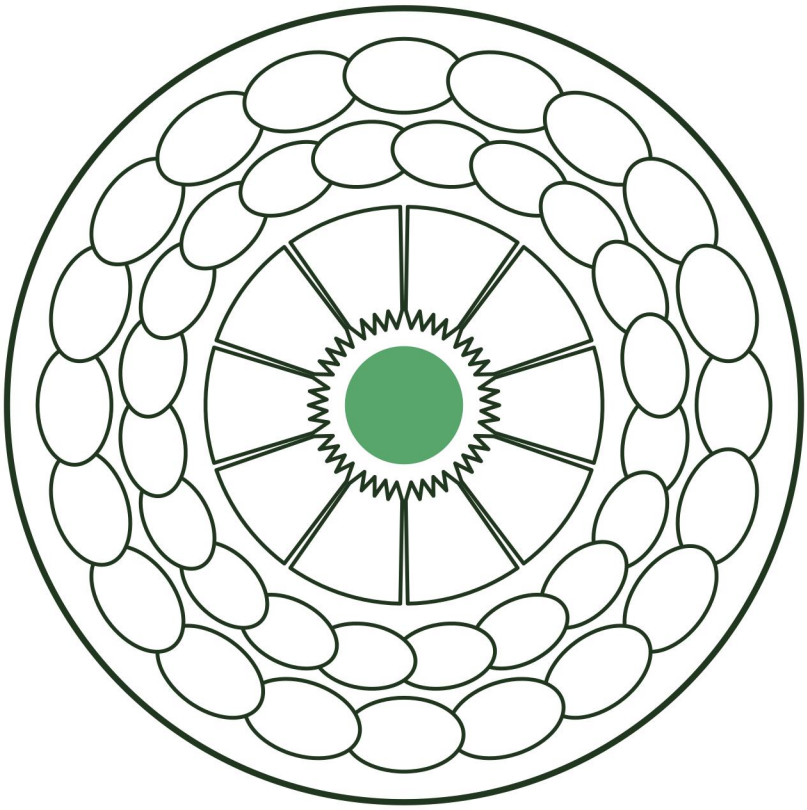
SOIL FOOD WEB FOUNDATION

Field Guide
to Soil Life



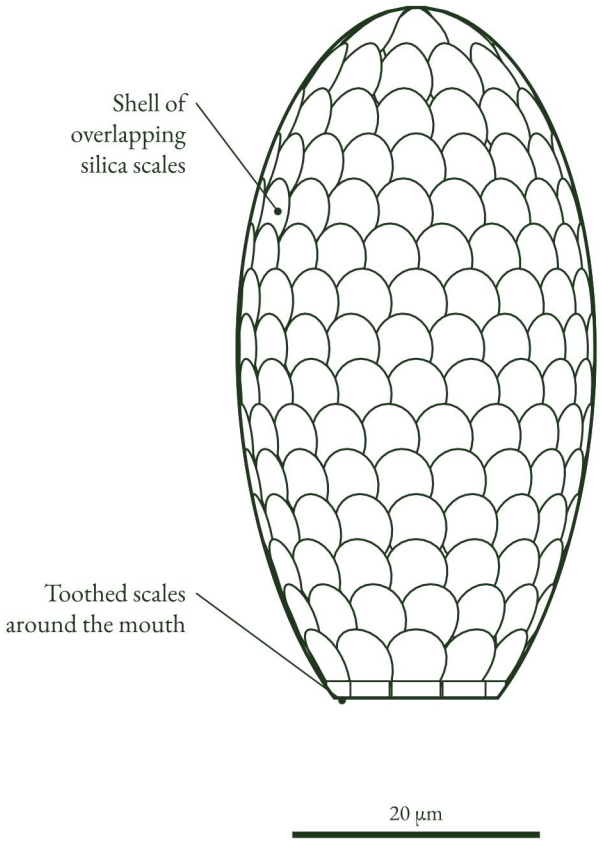
Eight organisms you will meet at the microscope

Testate amoebae · Nematodes · Fungi · Flagellates
Naked amoebae · Ciliates · Bacteria · Microarthropods

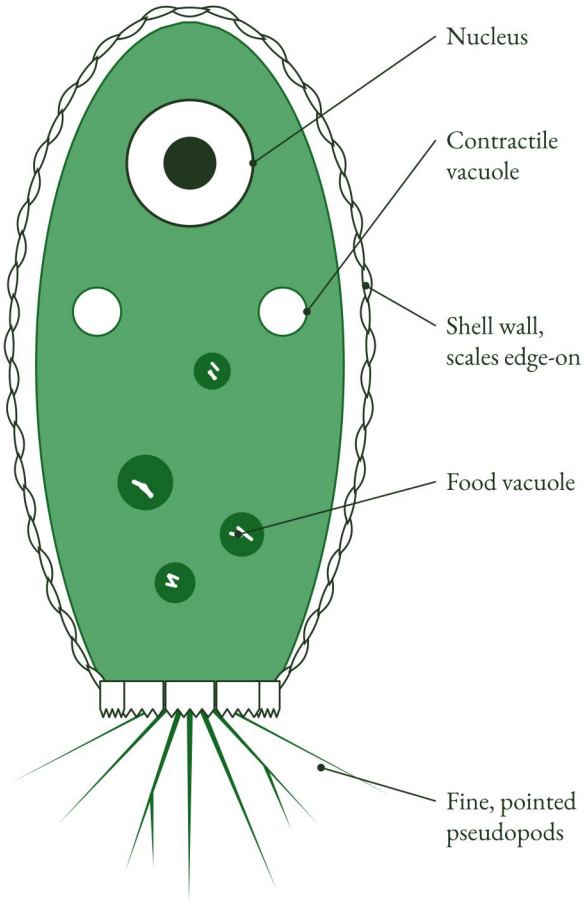
Euglypha

A shelled amoeba that builds its house from glass

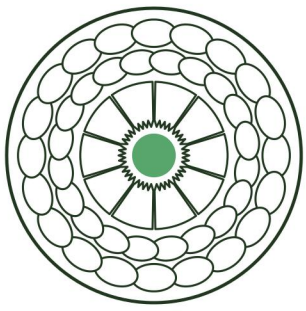
Testate amoeba · Rhizaria · Cercozoa · Euglyphida



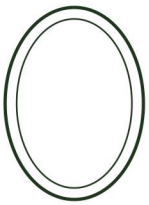
A Shell surface



B Living cell, mid focus



C The mouth, end-on



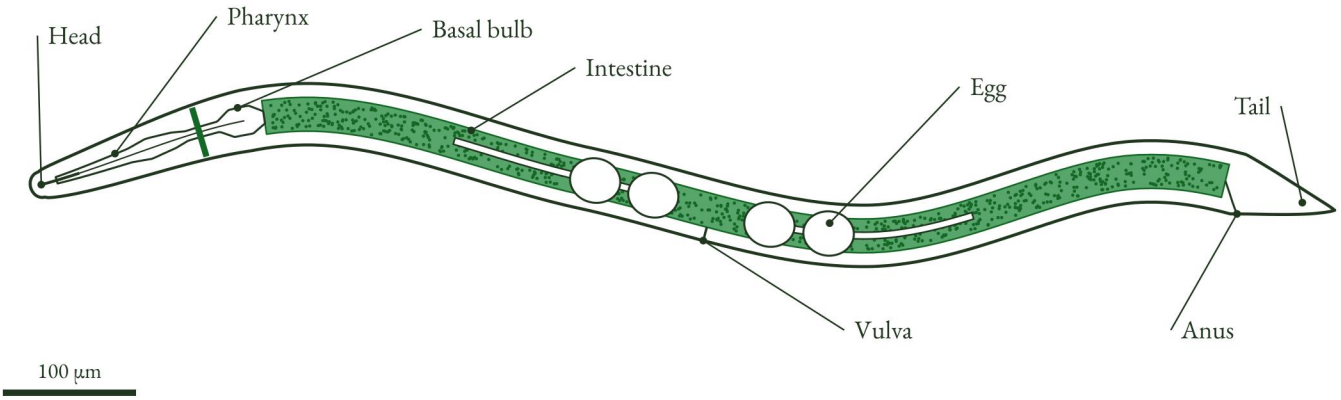
D One scale

Size	Shell about 20 to 100 µm long, depending on the species.
Where	Moist soil, leaf litter and moss.
Eats	Bacteria, fungi and small algae, caught with fine pointed pseudopods.
Shell	Silica scales the cell makes itself. It stores a full new set before it divides.
At 400x	A clear, glassy shell. Focus on the surface for the scale pattern, then on the middle for the cell. Look for the toothed mouth.

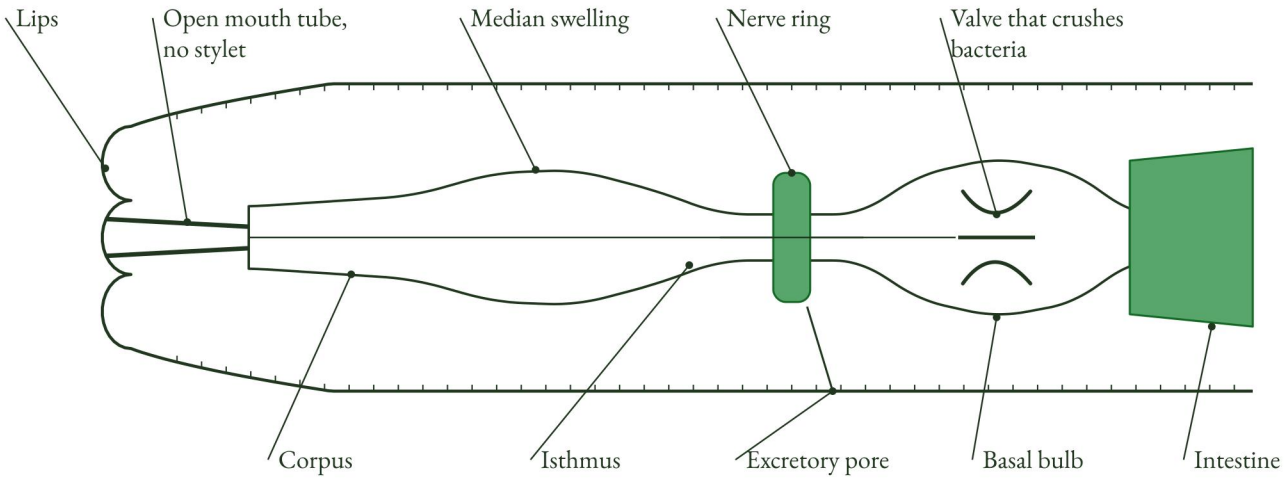
Bacterial-feeding nematode

A roundworm that grazes on bacteria and releases nitrogen for plants

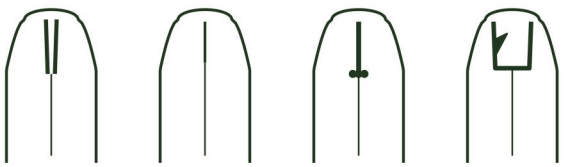
Nematoda · Rhabditida · for example Rhabditis, Cephalobus, Acrobeles



A Whole worm, female, about 1 mm



B Head and pharynx, where you identify it



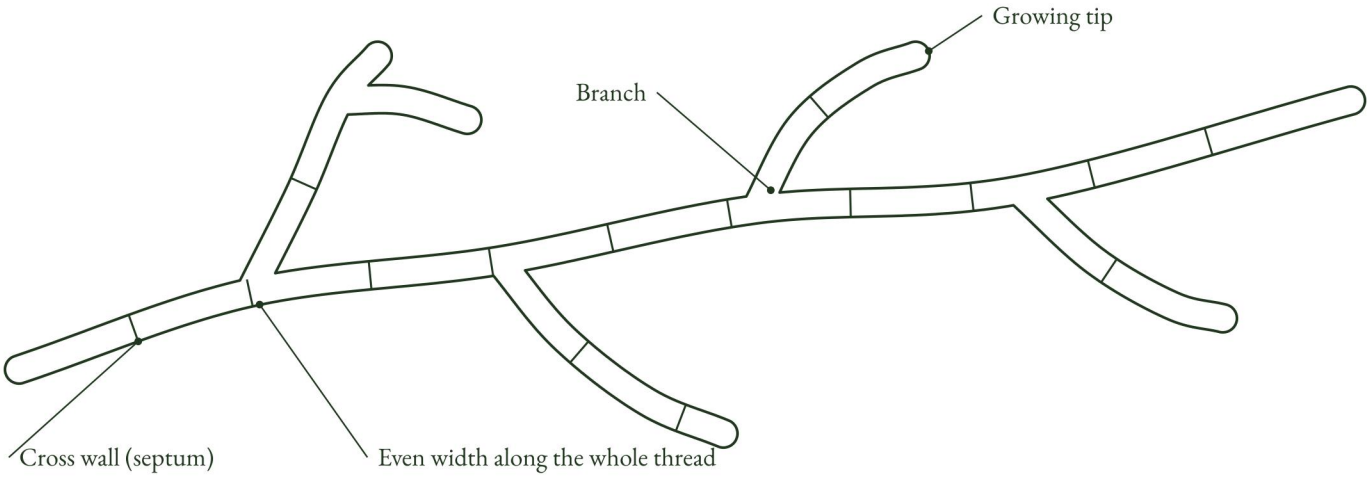
C What each group eats, read from the head

Size	Typically about 1 mm long and 50 µm wide.
Where	Water films around soil particles, around roots and in compost.
Eats	Bacteria, sucked in through the open mouth and crushed in the basal bulb.
Role	It eats more nitrogen than it needs and releases the rest as ammonium, a form plant roots can take up.
At 400x	Look at the head: an open tube, no stylet, and a round bulb that pumps as it feeds.

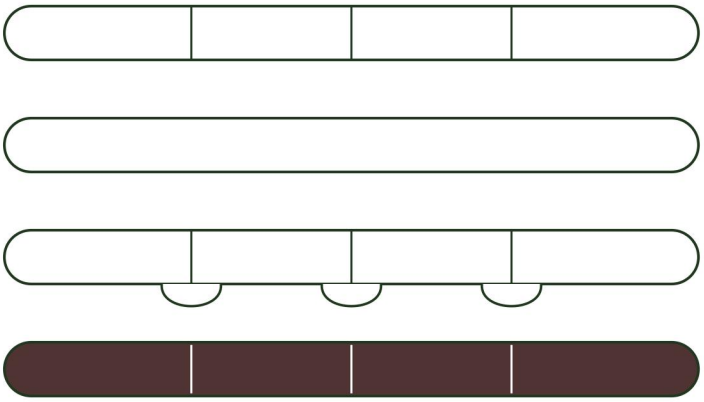
Fungal hyphae

The threads that hold soil together

Fungi · hyphae are the feeding threads, a network of them is mycelium



A A septate hypha, branching, about 400x



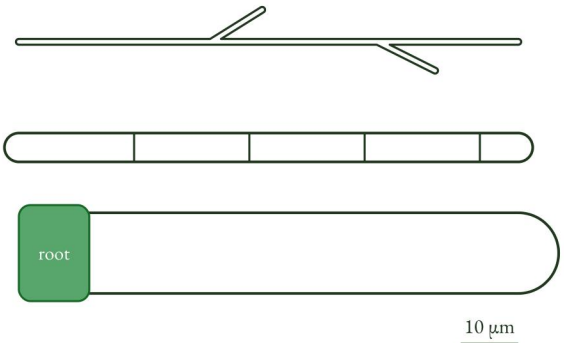
Septate
Cross walls at intervals. Most soil fungi.

Aseptate
No cross walls. Arbuscular mycorrhizal fungi, Mucor.

Clamp connection
A hook over each cross wall. Basidiomycetes.

Brown (pigmented)
Dark wall, often older or decay-resistant hyphae.

B Four kinds to tell apart



Actinobacteria
About 1 to 1.5 µm. Much thinner, branches at sharp angles.

Fungal hypha
Often 4 to 6 µm. Even width, cross walls or none.

Root hair
About 10 to 17 µm. Grows out of a root, no branches.

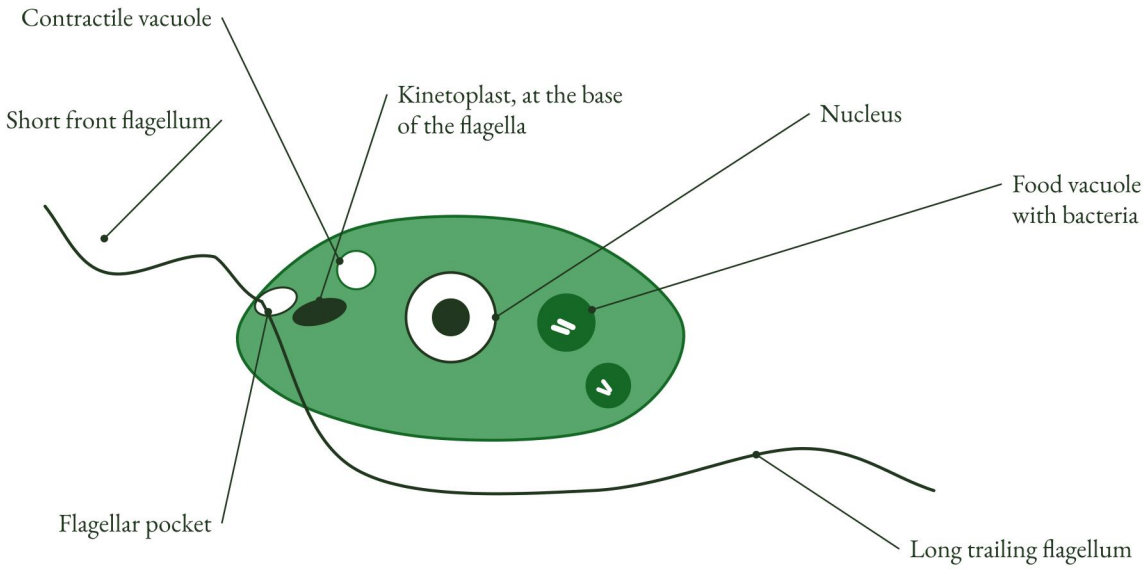
C Easy to mix up: all three at the same size

Eats	Dead plant material, even tough lignin and cellulose. Mycorrhizal fungi get sugars from roots.
Role	Binds soil crumbs together, holds nutrients in the soil, and reaches water and nutrients for roots.
At 400x	Look for an even-width tube with cross walls or none. Measure the width against a bacterium.

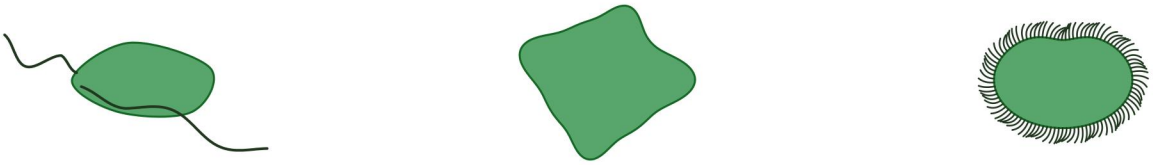
Flagellates

The smallest protozoa in soil

Protozoa · for example Bodo, Cercomonas, Spumella



A A Bodo-type flagellate, about 1000x



Flagellates
One or a few whips

Amoebae
Flowing pseudopods

Ciliates
Many short hairs

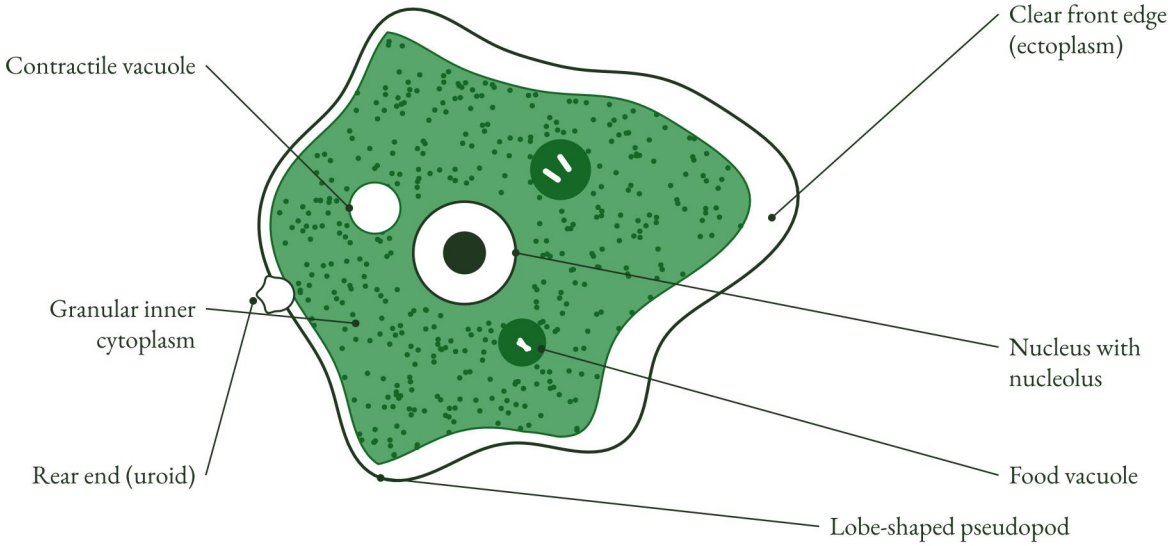
B The three protozoa groups, and how they move

Size	The smallest protozoa. Bodo saltans, a common one, is about 5 to 8 µm long.
Where	The water films around soil particles.
Eats	Bacteria. The short flagellum sweeps food in, the long one drives the cell.
Role	Bacteria hold more nitrogen than a flagellate needs, so it releases the rest as ammonium, a form plant roots can take up.
At 400x	Tiny cells that jerk and spin fast. The flagella are often too thin to see. Count them by their quick, twitching movement.

Naked amoebae

Shape-shifting cells that flow toward their food

Protozoa · Amoebozoa and others · without a shell



A A lobose amoeba, flowing to the right, about 400x



Limax
Slug-shaped, one pseudopod

Many lobes
Several pseudopods at once

Flat fan
Spread thin, wide front

With a shell
Testate amoeba, see No. 1

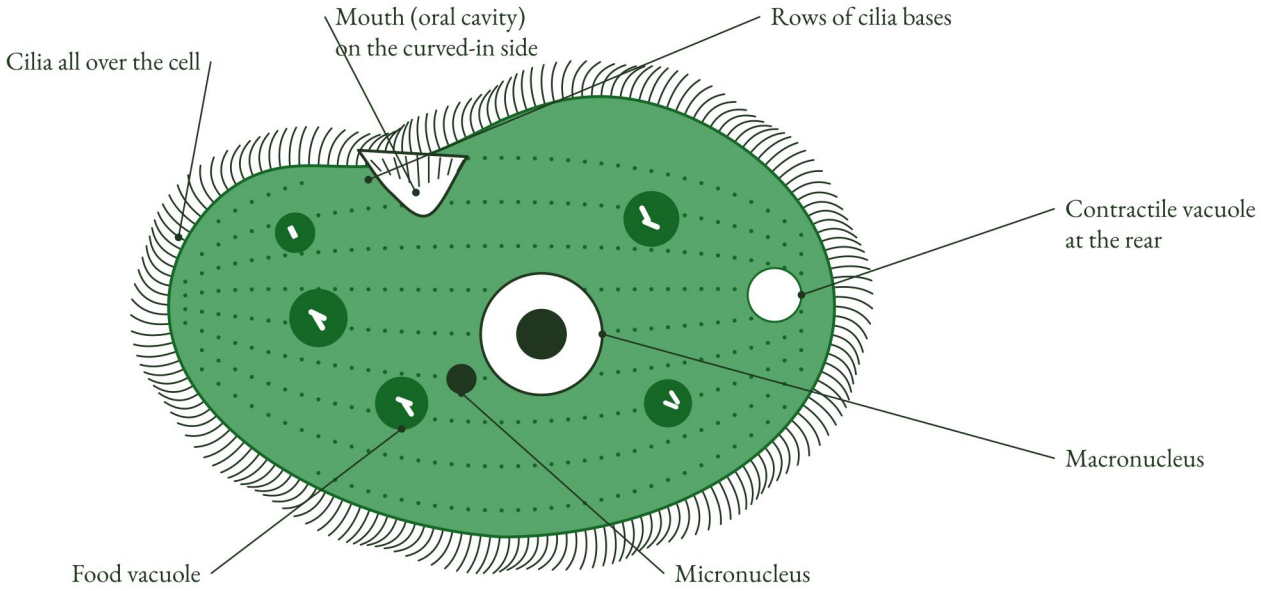
B Shapes you will meet in soil

Size	Between flagellates and ciliates in size. About 10,000 to 100,000 live in a gram of soil.
Where	Water films and the tiny pores inside soil crumbs, where other predators cannot reach the bacteria.
Eats	Mainly bacteria. It flows around its food and swallows it.
Role	Releases plant-available nitrogen, like the other protozoa.
At 400x	A cell that slowly changes shape. Watch for a clear cap at the front and grainy cytoplasm flowing in behind it.

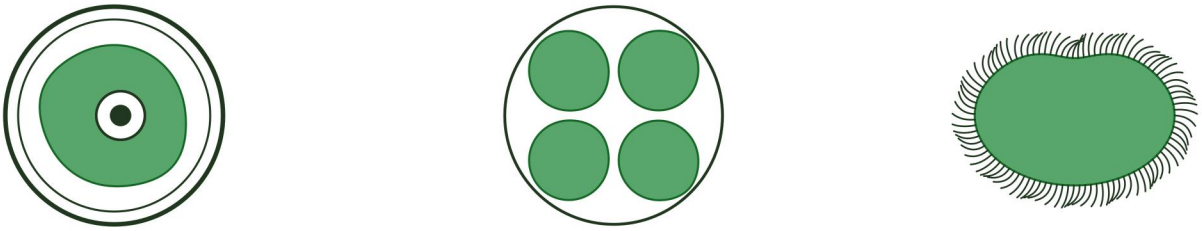
Ciliates

Fast swimmers covered in rows of tiny hairs

Protozoa · Ciliophora · for example Colpoda



A Colpoda, a common soil ciliate, about 400x



Resting cyst
Thick wall. How it survives dry soil.

Division cyst
Usually 4 new cells, sometimes 2 or 8.

Active cell
Swims when the soil is wet.

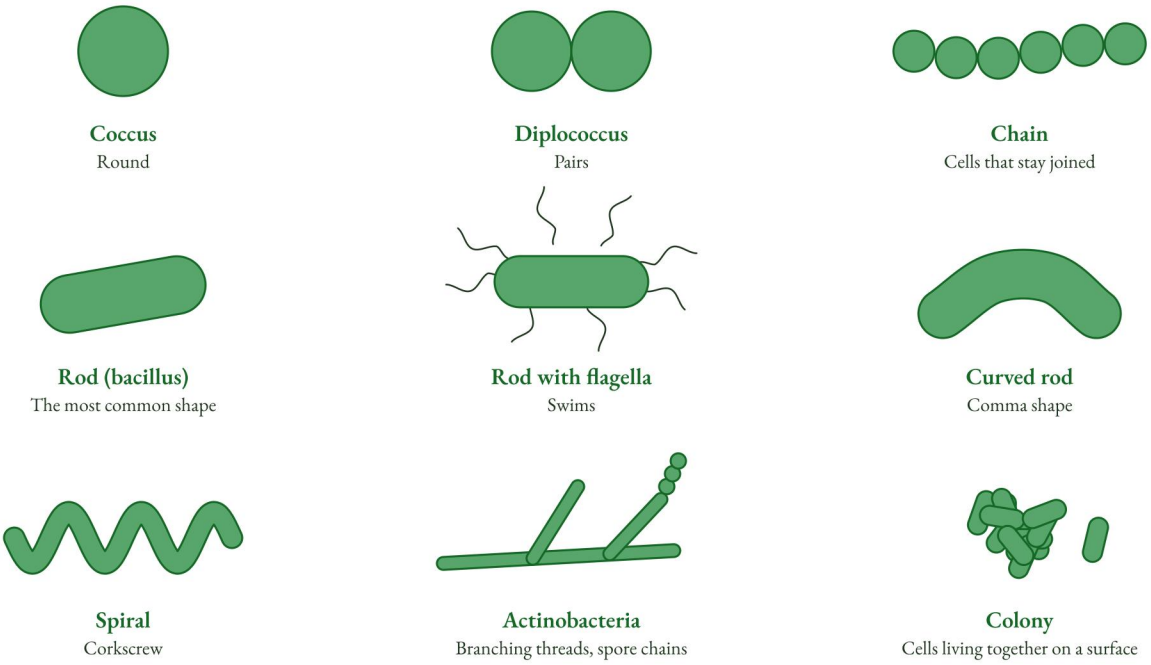
B Three stages of Colpoda

Size	Ciliates are the largest and least numerous protozoa. Some colpodeans, the group Colpoda belongs to, are as small as 10 µm.
Where	Water-filled pores and water films.
Eats	Bacteria, swept into the mouth by the cilia.
Role	Releases nitrogen like the other protozoa. In the Soil Food Web method, many ciliates point to soil that is short of oxygen.
At 400x	Fast, gliding swimmers that change direction often. Look for the fringe of cilia and the notch of the mouth.

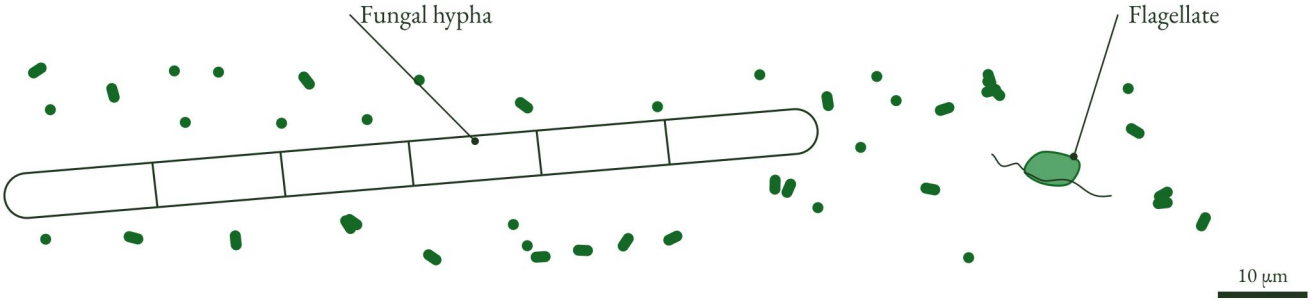
Bacteria

Tiny, countless, and the base of the soil food web

Bacteria and actinobacteria · single cells about 1 µm across



A Shapes, greatly enlarged



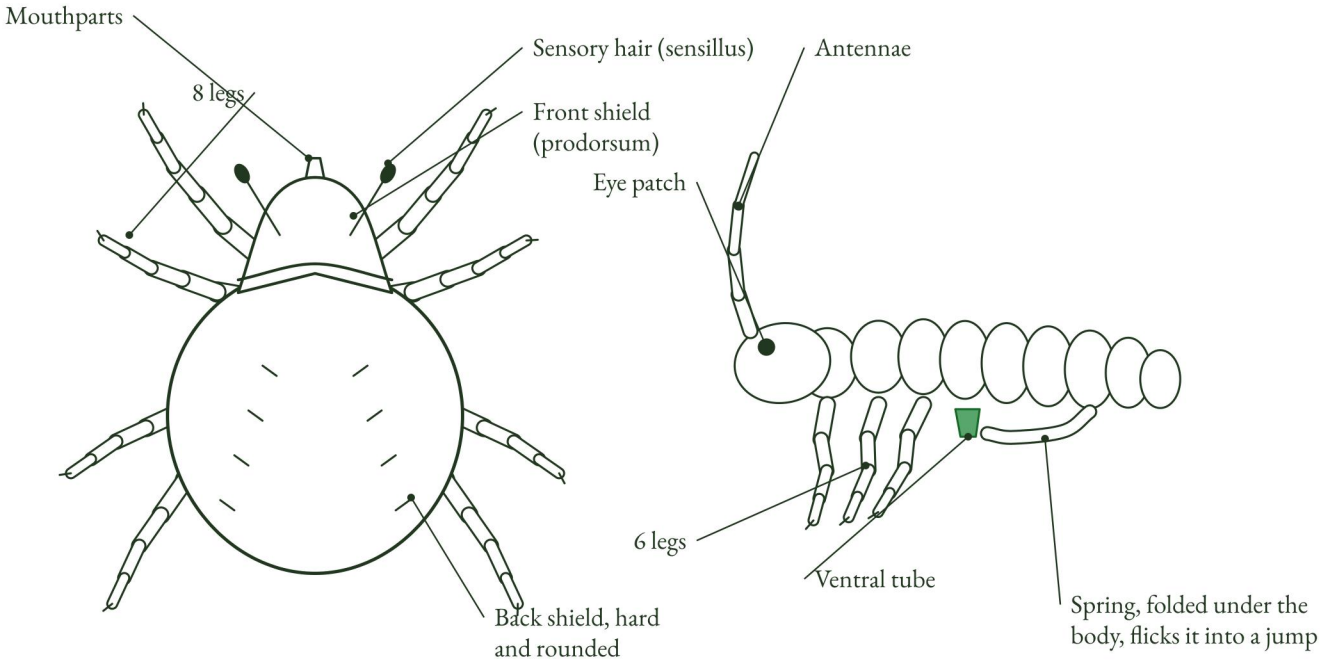
B At 400x: bacteria next to a fungal hypha and a flagellate, all at the same size

Size	About 0.5 to 2 µm. 100 million to 1 billion live in a teaspoon of productive soil.
Eats	Mostly simple sugars and other easy food from roots and decaying material.
Role	Make sticky substances that glue soil particles into tiny crumbs, hold nutrients, and feed the protozoa and nematodes that release those nutrients to plants.
At 400x	Dots and short rods, some moving. You count bacteria, you do not identify them.

Microarthropods

The shredders: mites and springtails

Arthropoda · oribatid mites (Acari) and springtails (Collembola)



A Oribatid mite, from above

B Springtail, from the side

Size	Oribatid mites 0.2 to 1 mm, springtails 0.2 to 5 mm. Many are just visible to the eye.
Where	Leaf litter, the top few centimetres of soil, and the larger pores.
Eats	Fungi, decaying plant material and bacteria. Some mites hunt nematodes and springtails.
Role	Shred litter so fungi and bacteria can reach it, spread spores, and release nutrients as they graze.
To tell	Mites have 8 legs and no antennae. Springtails have 6 legs, antennae, and jump.
To see	Use low power, 10x to 40x, or a stereo microscope. They are too big for 400x.

Field Guide to Soil Life

Every organism in this guide is drawn to show what you actually see at the microscope: the parts that tell you what it is, its true size, and what it does in the soil.

Each fact is checked against a published source. The sources are listed in the notes of each page.

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